ELECTRONICS AND COMMUNICATION ENGINEERING

# COMMUNICATION SYSTEM



## Baseband Transmission

Lecture No.: 4



By- SAKET VERMA SIR




1. Bandpass Sampling
2. Practice Questions




1. Quantization
2. Signal to Quantization Noise Ratio
3. Intersymbol Interference



# Bandpass Sampling

#Q. A band pass signal has significant frequency components in the range of 1.5 MHz to 2 MHz. If the signal is to be reconstructed from its samples, the minimum sampling frequency will be

A 1 MHz

B 2 MHz

C 3.5 MHz

D 4 MHz

1) $B = f_H - f_L = 2 - 1.5 = 0.5\ MHz.$

2) $K \longrightarrow \frac{f_H}{B} = \frac{2}{0.5} = 4$

$K = 4$

3) $(f_s)_{min} = \frac{2f_H}{K} = \frac{2 \times 2}{4} = 1\ MHz.$



# Bandpass Sampling

$\times f_s = 2f_m = 4\ MHz \times$

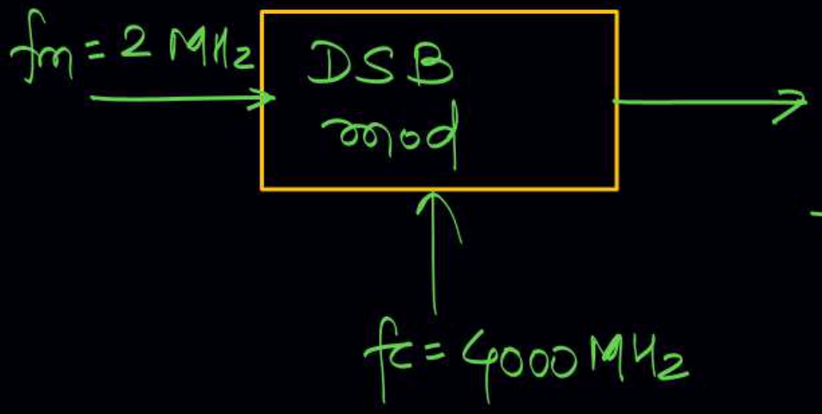
#Q. A 4 GHz carrier is DSB – SC modulated by a low pass message signal with maximum frequency of 2 MHz. The resultant signal is to be ideally sampled. The minimum frequency of the sampling impulse train would be

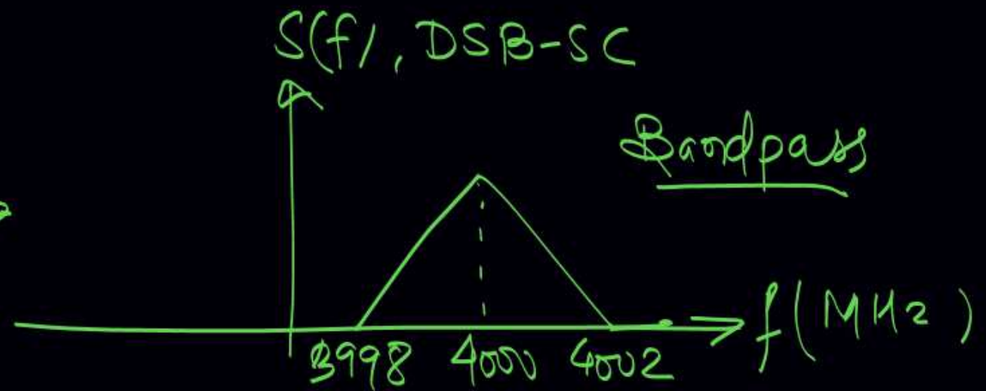
**A** 4 MHz

**B** 8 MHz

**C** 8 GHz

**D** 8.004 GHz




1) $B = 4$ MHz

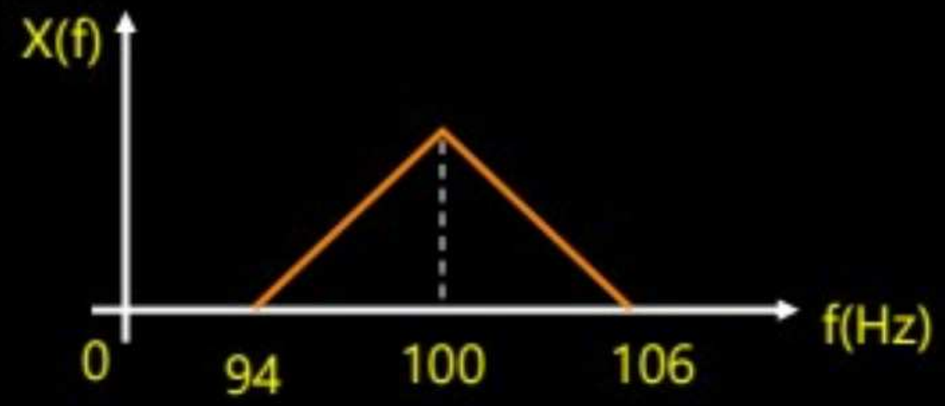
2) $K \rightarrow \frac{f_H}{B} = \frac{4002}{4} = 1000.5$

$K = 1000$

3) $(f_s)_{min} = \frac{2f_H}{K} = \frac{2 \times 4002}{1000} = 8.004$ MHz.

# Bandpass Sampling

#Q. Consider the sampling of the bandpass signal, whose spectrum is shown below. What minimum sampling frequency would you use from the options given below, so as to avoid aliasing?




A 24 Hz

B 26.5 Hz

C 48 Hz

D 212 Hz

1) $B = f_H - f_L = 106 - 94$

$= 12\ Hz$

2) $K \rightarrow \frac{f_H}{B} = \frac{106}{12} = 8.8$

$K = 8$

3) $(f_s)_{min} = \frac{2f_H}{K} = \frac{2 \times 106}{8} = 26.5\ Hz$

# Bandpass Sampling

#Q. For a signal with $f_c$ (Center frequency) = 1200 MHz and BW = 100 MHz. Which of the following sampling frequency $f_s$ will cause Spectrum Inversion?

A 287.5 MHz B 575 MHz

C 1150 MHz D 1600 MHz

$$\frac{2f_c+B}{m+1} < f_s < \frac{2f_c-B}{m}$$

$$\frac{2\times1200+100}{2} < f_s < \frac{2\times1200-100}{1}$$

$$1250 < f_s < 2300 \text{ MHz}$$

→ $m = 1$ odd

$m = 2$, even → no spectrum inversion.

$$\frac{2\times1200+100}{3} < f_s < \frac{2400-100}{2}$$

$$\frac{2500}{3} < f_s < \frac{2300}{2}$$

$$833 < f_s < 1150$$



# Bandpass Sampling

$f_c = 3\ kHz$ $B = 4\ kHz$

#Q. An analog signal has significant spectral components from 1 kHz to 5 kHz. What will be the sampling rate for this signal?

**A** 5 k samples/s **B** 4 k samples/s

**C** 8 k samples/s **D** 10 k samples/s

1) $B = 4\ kHz$

2) $K \rightarrow \frac{f_H}{B} = \frac{5}{4} = 1.25$

$K = 1$

3) $(f_s)_{min} = \frac{2f_H}{K} = \frac{2 \times 5}{1} = 10\ kHz$

# Bandpass Sampling

$x(t)\cos\omega_0 t \longrightarrow \frac{1}{2}\left[X(f-f_0)+X(f+f_0)\right]$

#Q. A signal x(t) has a bandwidth 2B about a carrier frequency of $f_c = 2$ GHz as shown in figure (a) below. In order to demodulate this signal, it is first mixed (Multiplied) with a local oscillator of frequency $f_{LO} = 1.5$ GHz, and then passed through an ideal LPF with a cut off frequency of 2.8 GHz. The output of the LPF is sent to a digitizer ADC with a sampling rate of 1.6 GHz as shown in figure (b) below. The maximum value of B so that the signal x(t) can be reconstructed from its samples according to Nyquist sampling theorem is 300 MHz

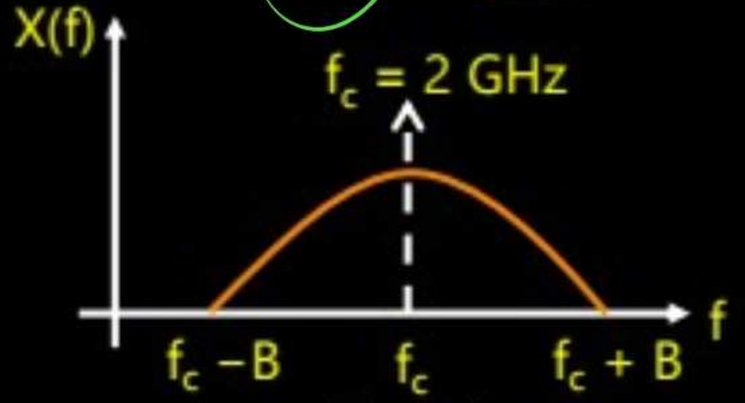


Fig (a)

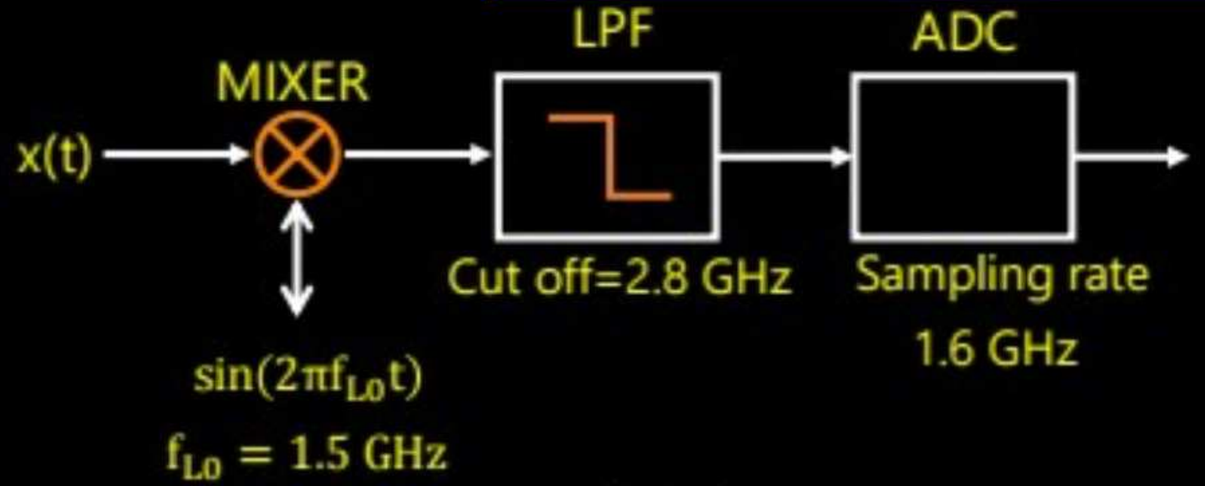
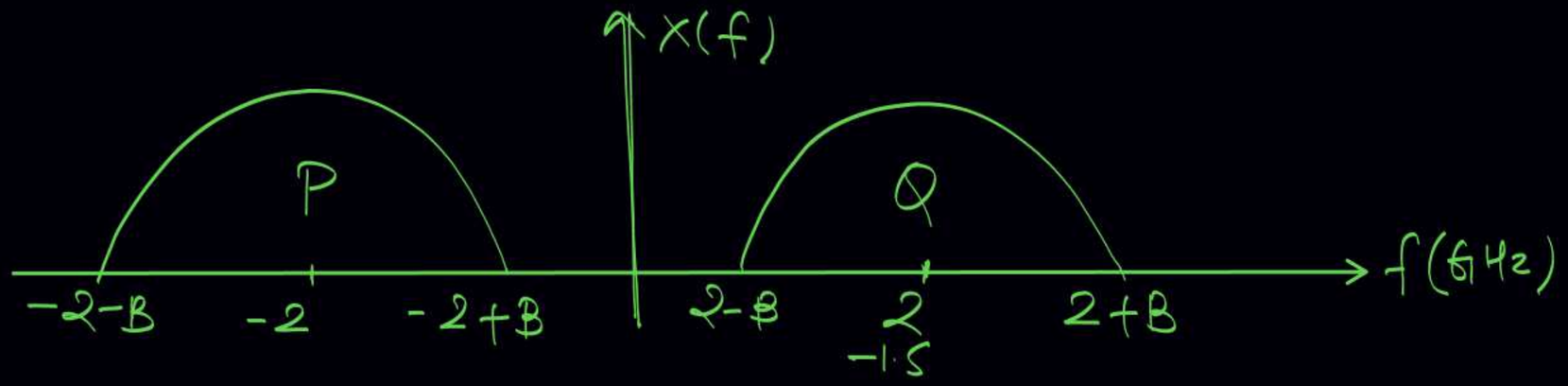
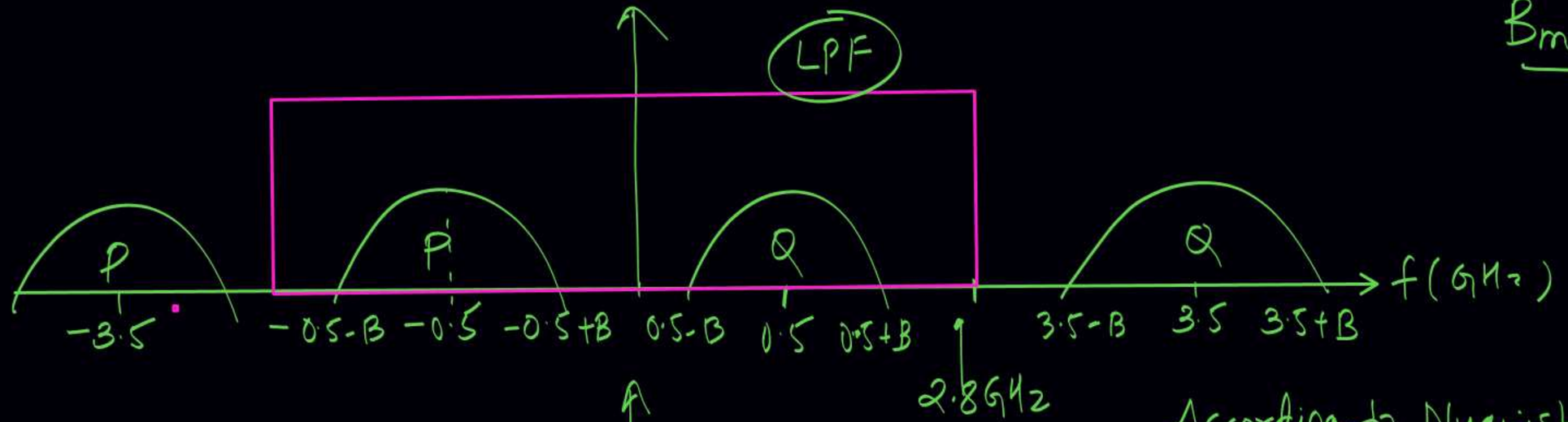
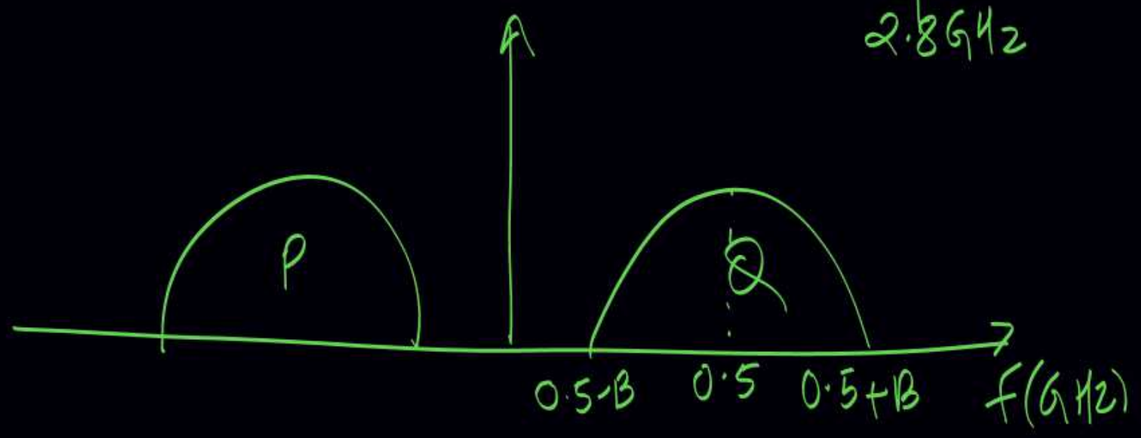


Fig (b)




$B_{max} = 300\ MHz$




According to Nyquist sampling.

$$f_s \geq 2f_H$$

$$1.6 \geq 2(0.5+B)$$

$$0.8 \geq 0.5+B$$

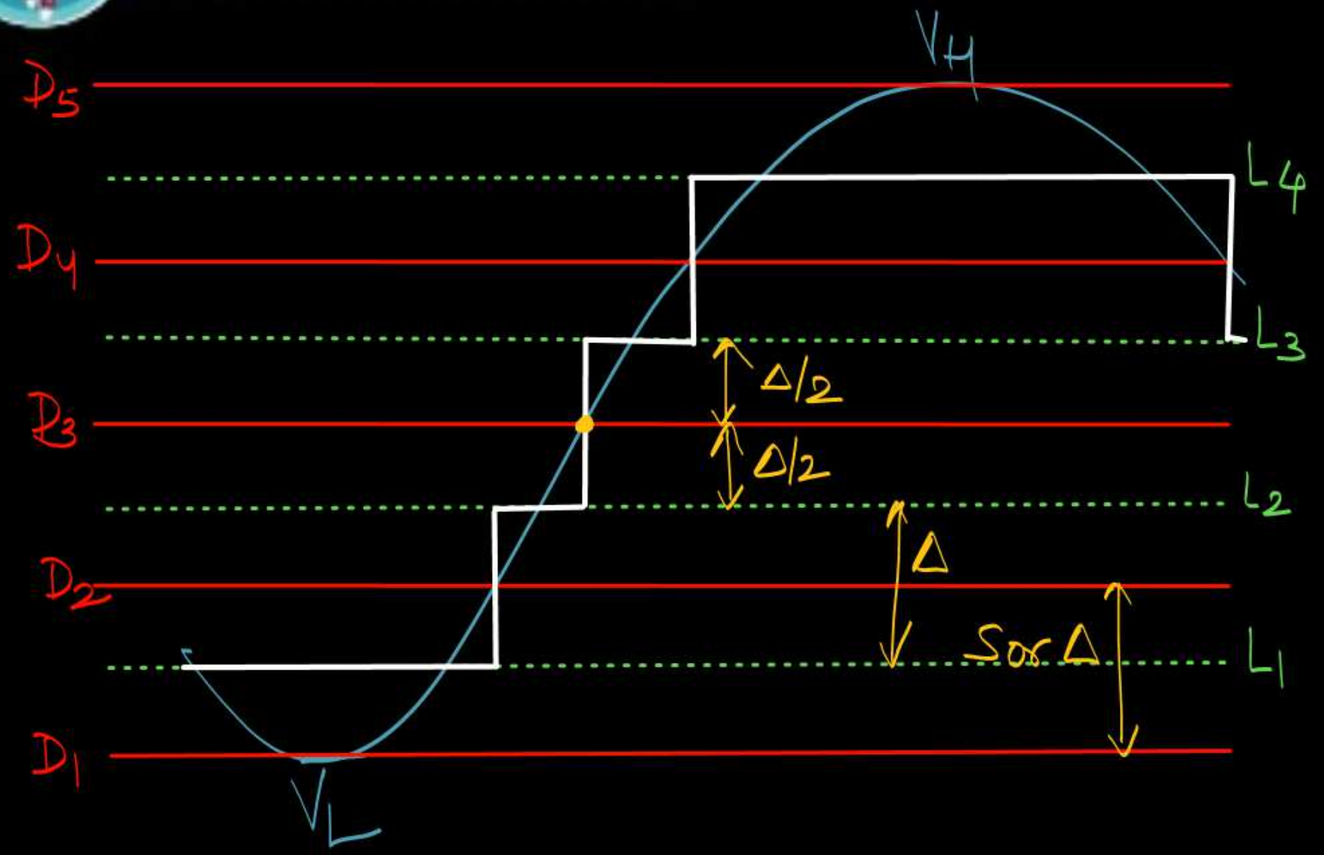
$$B < 0.3\ GHz$$



# Quantization

- The quantizing of an analog signal is done by discretizing the signal with several quantization levels.
- Quantization represents sampled values of the amplitude by a finite set of levels, which means converting a continuous-amplitude sample into a discrete-amp signal.

# Quantization




Decision Boundary.

$m(t)$

quantization levels or represent$^{n}$ levels.

o/p of quantizer (staircase).



# Quantization

- Quantization is many to one mapping
- The difference between m(t) and $m_q(t)$ is called quantization error.

$$q = |m(t) - m_q(t)|$$

- Difference between two quantization levels or decision boundaries is called step size.

$$\Delta = \frac{V_H - V_L}{L}$$

Where $(V_H - V_L)$ = Dynamic Range

$L = 2^n$ No. of Levels = L

n = No. of bits per sample $\Rightarrow$ $n = \log_2 L$



# Quantization

- The difference between an input value and its quantized value is called a Quantization Error.
- The quantization error $q$ has range $-\frac{\Delta}{2} \leq q \leq \frac{\Delta}{2}$
- The maximum error we could observe in quantization process is $\left|\pm\frac{\Delta}{2}\right| = \frac{\Delta}{2}$.



# Quantization

## Types of Quantization

- Depending on the step size Δ, there are two types of Quantization.
  - Uniform Quantization
  - Non-Uniform Quantization (Companding) → A-law, μ-law
- The quantization process in which step-size is fixed is called Uniform quantization.
  - Mid-rise Quantization
  - Mid-tread Quantization



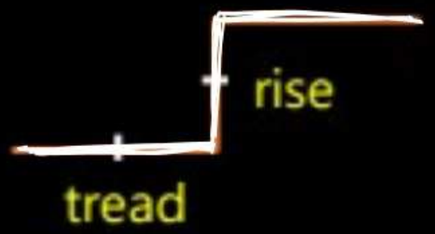
# Quantization

## Mid-Rise

The origin lies in the middle of a raising part of the stair-case like graph. The quantization levels in this type are even in number.

## Mid-Tread

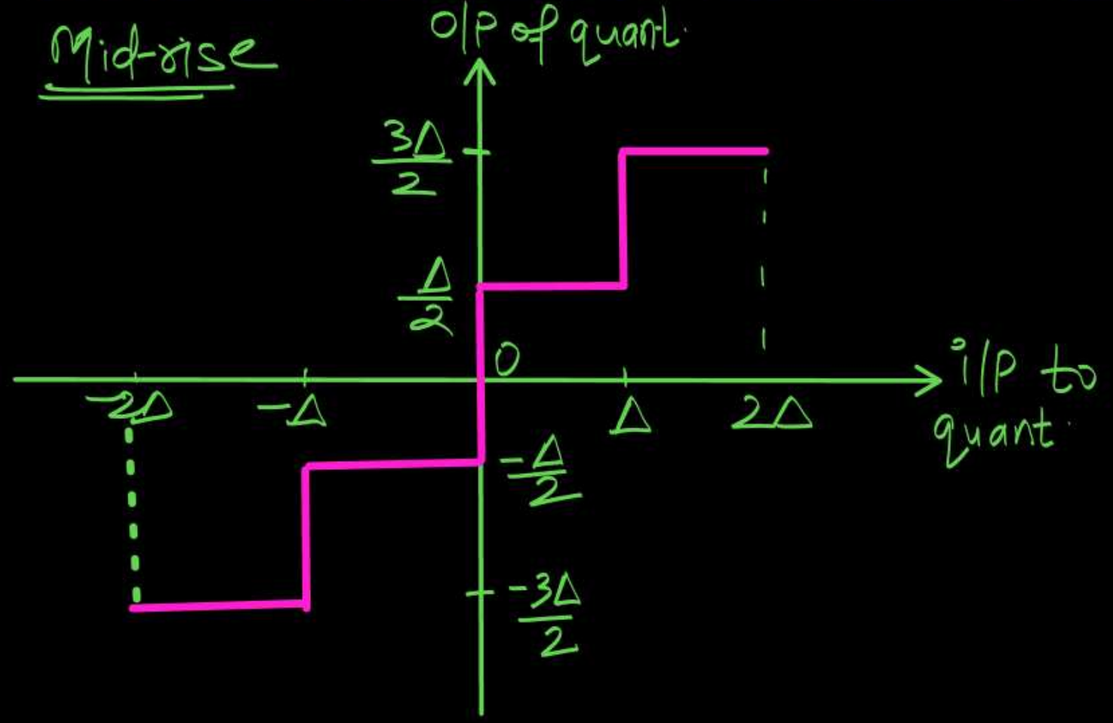
The origin lies in the middle of a tread of the stair-case like graph. The quantization levels in this type are odd in number.



# Quantization

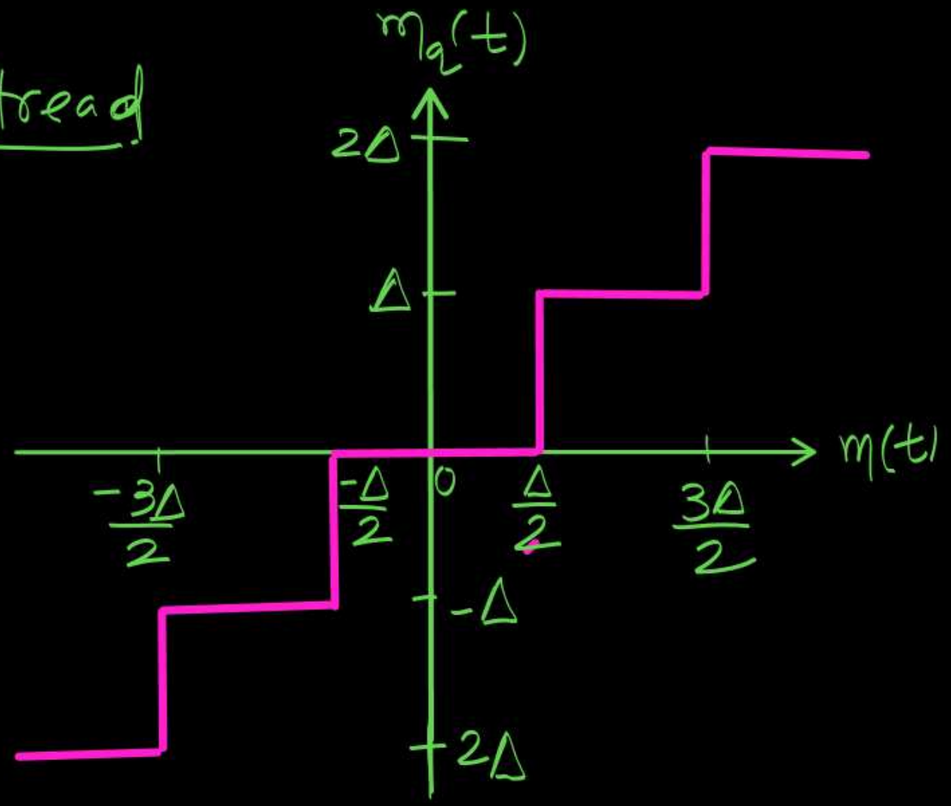
## Uniform Quantization Characteristics

Mid-rise




no. of quant. level $= 2^n$

Mid tread




no. of quant. level $= 2^n - 1$



# Encoder

- The process of assigning digital data bits to different quantization levels is called Encoding.
- If there are L number of levels, n bits are sufficient for encoding process, where $L = 2^n$.
- We have n-bits corresponding for each sample then we can write Bit Rate as -

$$R_b = n \times f_s \text{ (bits/sec)}$$

$$\frac{\text{no. of bit}}{\text{sample}} \times \frac{\text{sample}}{\text{sec.}} = \frac{\text{bit}}{\text{sec}}$$



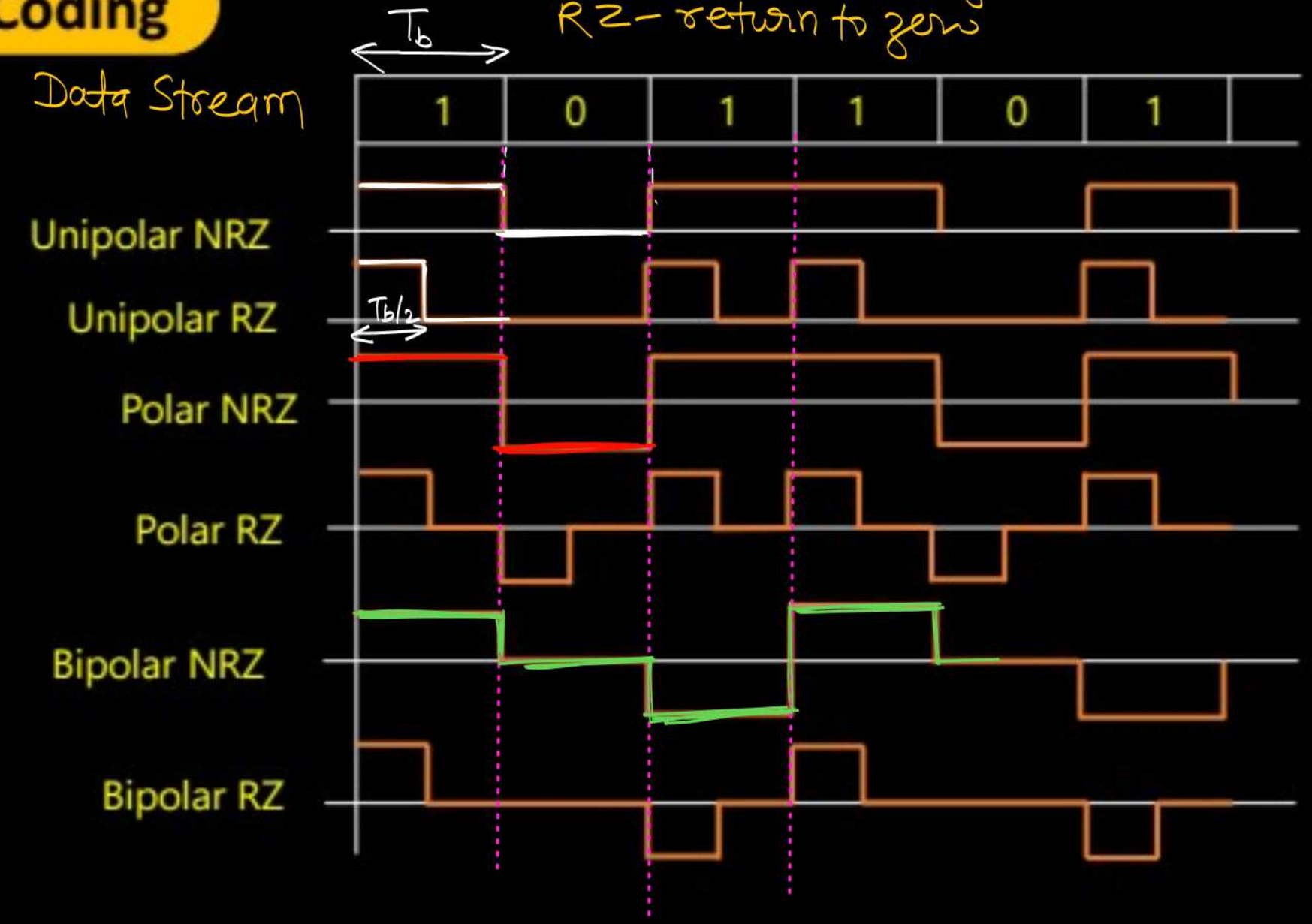
# Line Coding → Electrical representation of data.

- When there are only two levels High and Low, the encoding is called Line Encoding.
- The bit stream is represented in voltage levels High and Low.
- The three basic types of Line-Encoding are -

| Unipolar | Polar | Bipolar |
|---|---|---|
| Has two-voltage levels, (+) and (0). | Has two-voltage levels, (+) and (-). | Has three-voltage levels, (+), (-), and (0). |
| Positive pulse for 1, No-Pulse for 0. | Positive pulse for 1, Negative pulse for 0. | Positive and Negative Pulse alternate for 1, No-Pulse for 0. |

# Line Coding

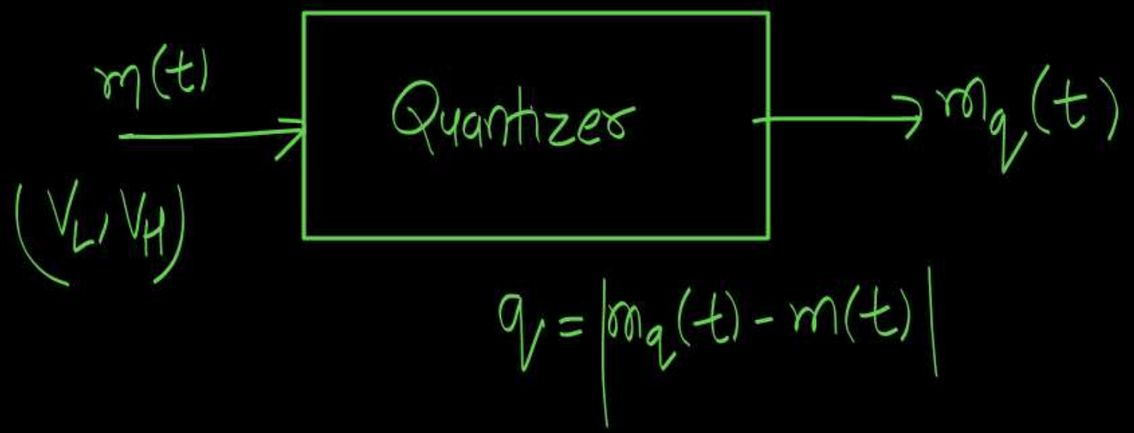
NRZ → non-return to zero
RZ - return to zero





# Signal to Quantization Noise Ratio

- The SQNR reflects the relationship between the maximum nominal signal strength and the quantization error (also known as quantization noise) introduced in the analog-to-digital conversion.
- The general expression of SQNR is given as $$\text{SQNR} = \frac{\text{Signal Power}}{\text{Noise Power}}$$



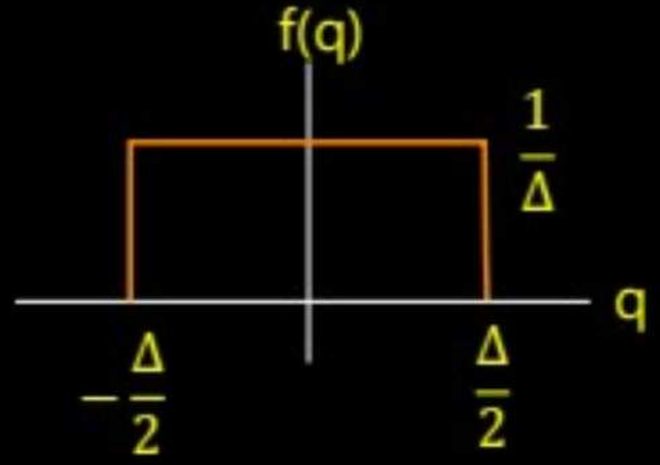
# Signal to Quantization Noise Ratio




- Consider the signal which is to be quantized is in range ($V_L$, $V_H$).

$$\text{Step Size } (\Delta) = \frac{V_H - V_L}{L}$$

- If step size is significantly small, then quantization error is assumed as uniformly distributed in range $\left(-\frac{\Delta}{2}, \frac{\Delta}{2}\right)$





# Signal to Quantization Noise Ratio

Mean square value of quantization error,

$$N_q = E(q^2) = \int_{-\infty}^{\infty} q^2 f(q)\, dq = \frac{b^2+ab+a^2}{3}$$

$$N_q = \frac{\left(\frac{\Delta}{2}\right)^2 + \left(-\frac{\Delta}{2}\times\frac{\Delta}{2}\right) + \left(\frac{-\Delta}{2}\right)^2}{3}$$

$$\boxed{N_q = \frac{\Delta^2}{12}}$$

$$N_q = \left(\frac{V_H - V_L}{L}\right)^2 \frac{1}{12} \Rightarrow \boxed{N_q \propto \frac{1}{L^2} \propto \frac{1}{2^{2n}}}$$

By increasing no. of bits or quantization levels, quant. noise power can be reduced.

$$N_q \propto \frac{1}{2^{2n}} \longrightarrow \text{corresponds to } n \text{ bits}$$

$$N_q' \propto \frac{1}{2^{2(n+1)}} \longrightarrow \text{corresponds to } n+1$$

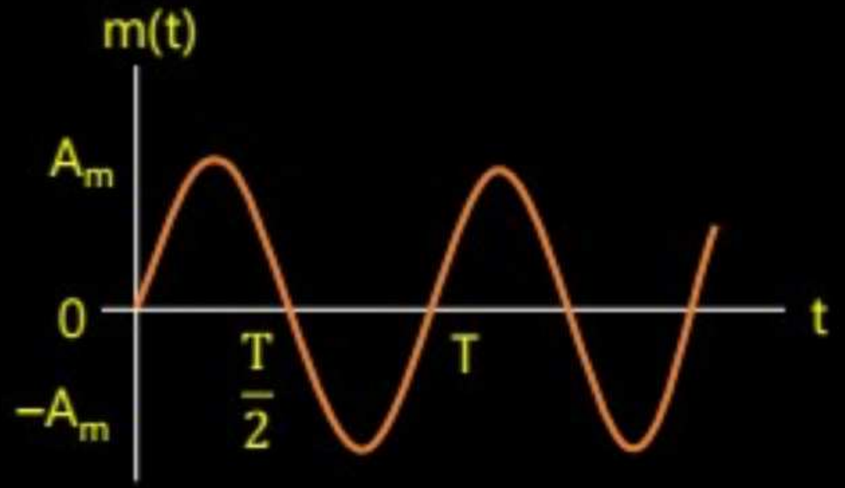
$$\frac{N_q'}{N_q} = \frac{1}{2^{2(n+1)}} \times \frac{2^{2n}}{1} = \frac{2^{2n}}{2^{2n} \times 2^2} = \frac{1}{4}$$

$$N_q' = \frac{N_q}{4}$$

# Signal to Quantization Noise Ratio

## Signal Power Calculation

- Assuming the message signal m(t) to be sinusoidal.




$V_H = A_m, \quad V_L = -A_m$

$\Delta = \frac{V_H - V_L}{L} = \frac{2A_m}{L}$

Signal power, $S_M = \frac{A_m^2}{2}$



# Signal to Quantization Noise Ratio

$$SQNR = \frac{S_M}{N_q} = \frac{\frac{A_m^2}{2}}{\frac{\Delta^2}{12}} = 6 \cdot \frac{A_m^2}{\left(\frac{2A_m}{L}\right)^2}$$

$$\boxed{SQNR = \frac{3}{2} L^2 = \frac{3}{2}(2)^{2n}}$$ for sinusoidal.

In general,

$SQNR \propto L^2$

$L^2$ ← square of quant: level

$$(SQNR)_{dB} = 10\log\left[\frac{3}{2} \times 2^{2n}\right] = 10\log\frac{3}{2} + 10\log 2^{2n}$$

$$\boxed{(SQNR)_{dB} = 1.76 + 6.02n}$$ for sinusoid.

$n \rightarrow n+1$ SQNR increase by 6dB

1 bit contributes 6dB in SQNR

# THANK you